Math 1019 — Definitions

A cumulative glossary, grouped by topic. Dates identify the lectures where the terms appear. Named laws and rules are included for reference. Wording and formulas have been clarified where the notes contain transcription errors. Coverage: September 11 through October 9, inclusive.

1. Statements and propositions

September 11 and 14

Statement An assertion expressed in words or mathematical notation. Mathematical relations such as $=$ and $>$ can serve the role of a verb. Whether an assertion is a proposition depends on whether it has a definite truth value.

Declarative sentence A sentence that makes an assertion, such as "The grass is wet." A declarative sentence is a proposition only if it is either true or false, but not both.

Imperative sentence A sentence that gives a command or instruction, such as "Hand in the assignment." It is not a proposition.

Interrogative sentence A sentence that asks a question, such as "Are you in this class?" It is not a proposition.

Exclamatory sentence A sentence that expresses strong feeling or emphasis, such as "Wow!" An exclamation alone is not a proposition; an exclamatory sentence that also makes a definite assertion can be one.

Proposition A declarative statement that is either true or false, but not both. Its truth value need not be known to us. For example, $2 + 3 = 5$ is a true proposition, and $2 + 3 = 6$ is a false proposition.

Truth value The value true (T) or false (F) assigned to a proposition.

Propositional variable A symbol, usually p, q, or r, that represents a proposition. For example, p might represent "It is raining."

Simple proposition A proposition treated as a single unit, without breaking it into propositions joined by logical connectives.

Compound proposition A proposition formed from other propositions using logical connectives, such as $\neg p$ or $p \wedge q$.

Logical connective An operation used to form a compound proposition. Its truth value is determined by the truth values of its inputs.

2. Logical connectives and truth tables

September 14 and 16

Negation (NOT) $\neg p$ means "not p." It is true exactly when p is false.

Conjunction (AND) $p \wedge q$ means "p and q." It is true exactly when both p and q are true.

Disjunction (inclusive OR) $p \vee q$ means "p or q." It is true when at least one of p and q is true, including when both are true. This is the default meaning of OR in propositional logic.

Exclusive OR (XOR) $p \oplus q$ is true exactly when one of p and q is true and the other is false.

NAND The negation of a conjunction, $\neg(p \wedge q)$. It is false exactly when both inputs are true. (Named in the September 14 notes.)

Implication (conditional) $p \to q$ means "if p, then q." It is false exactly when p is true and q is false; it is true in every other case. It rules out a true hypothesis with a false conclusion.

Hypothesis In $p \to q$, the proposition p. It is the condition assumed in the "if" part of the implication.

Conclusion In $p \to q$, the proposition q. It is the assertion in the "then" part. In an argument, the conclusion is the assertion that is meant to follow from the premises.

Biconditional (bi-implication) $p \leftrightarrow q$ means "p if and only if q," abbreviated "p iff q." It is true exactly when p and q have the same truth value. Equivalently, both $p \to q$ and $q \to p$ hold.

Truth table A table listing every combination of truth values of the propositional variables and the resulting values of expressions built from them. With n distinct variables, there are 2^n rows of assignments.

Operator precedence The order in which connectives are applied when parentheses do not specify the grouping. The order given in the notes is $\neg$, then $\wedge$, then $\vee$, then $\to$, then $\leftrightarrow$.

For quick reference:

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

3. Related implications

September 16

Relative to the original implication $p \to q$:

Converse $q \to p$. Exchange the hypothesis and conclusion.

Inverse $\neg p \to \neg q$. Negate both the hypothesis and conclusion.

Contrapositive $\neg q \to \neg p$. Exchange and negate both parts. An implication is logically equivalent to its contrapositive.

The converse and inverse are logically equivalent to each other, but neither is generally equivalent to the original implication.

4. Classification and logical equivalence

September 16, 18, and 21

Tautology A propositional expression that is true for every assignment of truth values to its variables. Example: $p \vee \neg p$.

Contradiction A propositional expression that is false for every assignment of truth values to its variables. Example: $p \wedge \neg p$.

Contingency A propositional expression that is true for at least one assignment and false for at least one assignment. Example: $p \to q$.

Satisfiable expression An expression for which at least one assignment of truth values makes it true. Both tautologies and contingencies are satisfiable; contradictions are not.

Logical equivalence Expressions S and T are logically equivalent, written $S \equiv T$, if they have the same truth value under every assignment. Equivalently, $S \leftrightarrow T$ is a tautology. Equivalent subexpressions may be substituted for one another.

4.1. Named logical laws

September 18; extensions on September 21 and 23

Here T and F denote the constant truth values true and false.

Identity laws $p \wedge T \equiv p$; $p \vee F \equiv p$.

Domination laws $p \vee T \equiv T$; $p \wedge F \equiv F$.

Idempotent laws (idempotency) $p \wedge p \equiv p$; $p \vee p \equiv p$.

Double negation law $\neg(\neg p) \equiv p$.

Negation laws $p \vee \neg p \equiv T$; $p \wedge \neg p \equiv F$.

Commutative laws The order of the operands can be exchanged: $p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$.

Associative laws The grouping can be changed: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$; $(p \vee q) \vee r \equiv p \vee (q \vee r)$.

Distributive laws Each of AND and OR distributes over the other:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Absorption laws $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$.

De Morgan's laws Negating a conjunction or disjunction negates each part and exchanges AND with OR:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

The same pattern applies to more than two propositions.

Conditional–disjunction equivalence (CD-equivalence) $(p \to q) \equiv (\neg p \vee q)$. Consequently, $\neg(p \to q) \equiv p \wedge \neg q$.

Contrapositive equivalence $(p \to q) \equiv (\neg q \to \neg p)$.

Biconditional equivalence $(p \leftrightarrow q) \equiv ((p \to q) \wedge (q \to p))$.

5. Predicates and quantifiers

September 21 and 23

Domain (of discourse) The collection of values a variable is allowed to take. Values must make the propositional function meaningful. For example, one possible domain for $x > 5$ is the integers.

Propositional function An expression such as $P(x)$ whose truth value depends on its variable. Substituting a particular value from the domain produces a proposition. For example, if $P(x)$ is $x + 7 = 13$, then $P(6)$ is true and $P(5)$ is false.

Subject The object or variable about which a property is asserted. In the September 21 example $Q(x)$, the subject is x.

Predicate The property asserted of the subject, such as "is greater than 5." The notation $P(x)$ is also commonly called a predicate; predicates may involve more than one variable, as in $P(x, y)$.

Quantifier A symbol specifying how widely a predicate holds over a domain. The quantifiers introduced in the notes are $\forall$ and $\exists$.

Universal quantification $\forall x P(x)$ means "$P(x)$ is true for every x in the domain." It is false if there is even one domain element for which $P(x)$ is false.

Existential quantification $\exists x P(x)$ means "there is at least one x in the domain for which $P(x)$ is true." It does not mean exactly one.

Unique existential quantification $\exists! x P(x)$ means "there exists exactly one x in the domain for which $P(x)$ is true." It asserts both existence and uniqueness. (October 7.)

Counterexample An example that disproves a claim. For $\forall x P(x)$, it is a value a in the domain for which $P(a)$ is false.

Finite domain A domain with finitely many elements. For a domain consisting of $x_1, ..., x_n$, quantification can be expanded as

$$\forall x P(x) \equiv P(x_1) \wedge ... \wedge P(x_n)$$

$$\exists x P(x) \equiv P(x_1) \vee ... \vee P(x_n)$$

Negation of quantifiers (De Morgan's laws for quantifiers) Negating a quantified statement exchanges the quantifier and negates the predicate:

$$\neg(\forall x P(x)) \equiv \exists x(\neg P(x))$$

$$\neg(\exists x P(x)) \equiv \forall x(\neg P(x))$$

Nested quantifiers Multiple quantifiers applied to a predicate involving multiple variables. Read them from left to right. For example, $\forall x \exists y P(x, y)$ allows the choice of y to depend on x, whereas $\exists y \forall x P(x, y)$ requires one y that works for every x.

Commuting quantifiers Quantifiers commute if exchanging their order preserves the meaning. Two universal quantifiers commute, as do two existential quantifiers:

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

A universal and an existential quantifier generally cannot be exchanged.

Mixed-quantifier implication A single witness that works for every element also supplies a witness for each element separately:

$$(\exists x \forall y P(x, y)) \rightarrow (\forall y \exists x P(x, y)).$$

The converse does not hold in general. (September 23.)

Implicit universal quantifier A universal quantifier understood from context rather than written explicitly. For example, "If m and n are even integers, then $m + n$ is even" is a claim about all integers m and n. (September 25.)

5.1. Application from the notes: continuity

September 23

Continuity at a point A function $f : \mathbb{R} \to \mathbb{R}$ is continuous at x_0 if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, (|x - x_0| < \delta \to |f(x) - f(x_0)| < \varepsilon).$$

Every positive error bound ε permits a positive distance bound δ that works for all x that close to x_0. Its negation is

$$\exists \varepsilon > 0, \forall \delta > 0, \exists x \in \mathbb{R}, (|x - x_0| < \delta \wedge |f(x) - f(x_0)| \geq \varepsilon).$$

6. Arguments and rules of inference

September 23 and 25; extensions on September 28 and 30

Argument A collection of premises together with a proposed conclusion. The symbol $\therefore$ is read "therefore" and introduces the conclusion.

Premises (hypotheses) The propositions assumed or given in an argument, usually written $p_1, p_2, ..., p_n$.

Valid argument An argument in which the premises cannot all be true while the conclusion is false. With premises $p_1, ..., p_n$ and conclusion q, validity means

$$(p_1 \wedge ... \wedge p_n) \to q$$

is a tautology. Validity concerns the inference; it does not assert that the premises are actually true.

Rule of inference A valid argument pattern that can be used to derive a conclusion from known or assumed premises.

Fallacy An invalid argument pattern. To show invalidity, give an assignment making every premise true and the conclusion false.

6.1. Rules used or named in the notes

Simplification From $p \wedge q$, infer p (or infer q).

Addition From p, infer $p \vee q$.

Conjunction rule From p and q as separate premises, infer $p \wedge q$.

Modus ponens From $p \to q$ and p, infer q.

Hypothetical syllogism From $p \to q$ and $q \to r$, infer $p \to r$.

Disjunctive syllogism From $p \vee q$ and $\neg p$, infer q. Equivalently, from $p \vee q$ and $\neg q$, infer p. (September 30.)

Resolution From $p \vee q$ and $\neg p \vee r$, infer $q \vee r$. In the September 25 example, the remaining terms are both q, giving $q \vee q$, which is equivalent to q.

Universal instantiation From $\forall x P(x)$, infer $P(a)$ for any particular element a of the domain.

Universal generalization If $P(a)$ has been proved for an arbitrary element a of the domain, infer $\forall x P(x)$. The proof must not rely on assumptions specific to that element. (September 28.)

Existential instantiation From $\exists x P(x)$, introduce a new symbol a for an element satisfying $P(a)$. This element is a witness, not an arbitrary element. Use separate new symbols for independent existential assertions; their witnesses need not be the same. (September 28.)

Existential generalization From $P(a)$ for a particular element a of the domain, infer $\exists x P(x)$. The element a supplies an example showing that the existential statement is true. (September 28.)

Universal modus ponens From $\forall x(P(x) \to Q(x))$ and $P(a)$, infer $Q(a)$, where a belongs to the domain.

Universal modus tollens From $\forall x(P(x) \to Q(x))$ and $\neg Q(a)$, infer $\neg P(a)$, where a belongs to the domain.

6.2. Common fallacies

Affirming the conclusion Incorrectly inferring p from $p \to q$ and q. Counterexample: p is false and q is true.

Denying the hypothesis Incorrectly inferring $\neg q$ from $p \to q$ and $\neg p$. Counterexample: p is false and q is true.

Affirming the disjunction Incorrectly inferring $\neg q$ from $p \lor q$ and p (or inferring $\neg p$ from $p \lor q$ and q). Counterexample: both p and q are true. Inclusive OR allows both.

7. Proof terminology and parity

September 25 and 28

Proof A valid mathematical argument establishing that a statement follows from the assumptions, definitions, and results being used.

Theorem A mathematical statement that has been proved, usually one regarded as important.

Proposition (as a proved result) A neutral name for a mathematical result that has been proved. This use differs from "proposition" in logic, where the statement may be true or false and need not be proved.

Lemma A proved result used as a stepping stone toward another result.

Corollary A proved result that follows readily from an established result.

Conjecture A statement proposed as true but not yet proved or disproved.

Even integer An integer n is even if and only if there exists an integer k such that $n = 2k$:

$$\exists k \in \mathbb{Z}, n = 2k.$$

Odd integer An integer n is odd if and only if there exists an integer k such that $n = 2k + 1$:

$$\exists k \in \mathbb{Z}, n = 2k + 1.$$

8. Proof techniques

September 28 and 30; vacuous proof on October 2

Direct proof To prove $p \to q$, assume p and use definitions, known facts, and rules of inference to establish q. For a universally quantified claim, keep the elements under consideration arbitrary.

Proof by contraposition Prove $p \to q$ by proving the logically equivalent implication $\neg q \to \neg p$. This can be easier when the negation of the conclusion gives a more useful starting assumption.

Proof by contradiction To prove a statement p, assume $\neg p$ and derive a contradiction. Since the assumption leads to something impossible, conclude that p is true.

Proof by cases (case analysis) Establish a conclusion q separately under each of the cases $p_1, \ldots, p_n$, and show that the cases cover every possibility:

$$p_1 \vee \ldots \vee p_n.$$

Once every implication $p_i \to q$ is proved, conclude q. (September 30.)

Exhaustive cases Cases whose disjunction is guaranteed to hold, so no possibility is left out. For example, the remainder when an integer is divided by 3 must be 0, 1, or 2. Cases need not be mutually exclusive. (September 30.)

Proof of existence A proof of $\exists x P(x)$. Giving an element a of the domain and showing $P(a)$ proves the claim by existential generalization.

Existence by cases An existence proof using exhaustive alternatives, with a suitable example in every case. It can establish that an example exists without determining which case actually holds. In the September 30 example, either $\sqrt{2}^{\sqrt{2}}$ is rational, or it is irrational; each case gives irrational x and y for which x^y is rational.

Vacuous proof A proof of an implication by showing that its hypothesis is always false. For example, $\forall x(x \in \varnothing \to x \in S)$ is true because no x belongs to the empty set. (October 2.)

9. Numbers and divisibility

September 28 and 30; number-set notation on October 2

Rational number A real number r that can be written $r = \frac{p}{q}$ for integers p and q with $q \neq 0$. Equivalently, the denominator can be required to be a positive integer.

Irrational number A real number that is not rational; it cannot be expressed as a quotient of integers with a nonzero denominator. For example, $\sqrt{2}$ is irrational.

Multiple and divisibility For integers a and b, the integer b is a multiple of a if $b = ak$ for some integer k. In this case, a divides b. For example, being even means being a multiple of 2. (September 30.)

Quotient and remainder When an integer n is divided by a positive integer d, it can be written uniquely as $n = dk + r$, where k and r are integers and $0 \leq r < d$. Here k is the quotient and r is the remainder. The possible remainders give exhaustive cases for a proof. (September 28 and 30.)

9.1. Common sets of numbers

October 2

Natural numbers $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$. In this course, 0 is a natural number.

Integers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

Positive integers $\mathbb{Z}^+ = \{x \in \mathbb{Z} \mid x > 0\} = \{1, 2, 3, \ldots\}$.

Rational numbers $\mathbb{Q}$, the set of all rational numbers:

$$\mathbb{Q} = \left\{ x \in \mathbb{R} \mid \exists p \in \mathbb{Z}, \exists q \in \mathbb{Z}^+, x = \frac{p}{q} \right\}.$$

Positive rational numbers $\mathbb{Q}^+ = \{x \in \mathbb{Q} \mid x > 0\}$.

Real numbers $\mathbb{R}$, the numbers on the real number line, including both rational and irrational numbers.

Complex numbers $\mathbb{C} = \{a + bi \mid a \in \mathbb{R} \wedge b \in \mathbb{R}\}$, where $i^2 = -1$.

9.2. Intervals

October 2

Interval A subset of $\mathbb{R}$ that contains every real number between any two of its elements. Parentheses exclude an endpoint; square brackets include it.

Open interval $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.

Closed interval $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.

Half-open intervals $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$ and $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$.

10. Sets and membership

October 2

Set A collection of distinct objects, called its elements. A set is determined by which elements it contains: changing their order or repeating them in a description does not change the set.

Element (membership) $x \in A$ means that x is an element of the set A; $x \notin A$ means that x is not an element of A.

Roster notation Describe a set by listing its elements between curly brackets, separated by commas. For example, $A = \{1, 4, 7, 9\}$. An ellipsis may indicate a clear pattern, as in $\{4, 5, 6, ..., 28\}$.

Set-builder notation Describe a set by a condition its elements must satisfy: $\{x \mid P(x)\}$, or $\{x \in D \mid P(x)\}$ when restricting x to a domain D. The vertical bar is read "such that."

Set equality Sets S and T are equal if and only if they have the same elements:

$$S = T \leftrightarrow \forall x(x \in S \leftrightarrow x \in T).$$

To show that two sets are unequal, find an element in one but not the other.

Empty set The unique set with no elements, written $\varnothing$ or $\{\}$. The set $\{\varnothing\}$ is not empty: its one element is the empty set.

Subset $A \subseteq B$ means that every element of A is also an element of B:

$$\forall x(x \in A \rightarrow x \in B).$$

Every set is a subset of itself, and $\varnothing$ is a subset of every set.

Superset $B \supseteq A$ means $A \subseteq B$; that is, B contains every element of A.

Strict subset (proper subset) $A \subset B$ means that $A \subseteq B$ and $A \neq B$. Thus B has at least one element that is not in A.

Not a subset $A \not\subseteq B$ means that some element of A does not belong to B:

$$\exists x(x \in A \wedge x \notin B).$$

11. Set size and constructions

October 2

Finite set A set whose elements can all be counted, with the count equal to a nonnegative integer n.

Cardinality The number of distinct elements of a finite set S, written $|S|$. For example, $|\varnothing| = 0$ and $|\{\varnothing\}| = 1$. A set that is itself an element counts as one element: the set $\{\varnothing, 1, \{1, 2, 4\}\}$ has cardinality 3.

Power set The set of all subsets of a set S, written $P(S)$:

$$A \in P(S) \leftrightarrow A \subseteq S.$$

For example, $P(\{a, b\}) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$. If S is finite, then $|P(S)| = 2^{|S|}$.

Ordered pair A pair (a, b) in which position matters. Two ordered pairs are equal exactly when their corresponding entries are equal: $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Ordered n-tuple A list $(x_1, x_2, ..., x_n)$ with n entries in specified positions. Equality requires equality in every corresponding position.

Cartesian product The set of ordered pairs formed by choosing the first entry from A and the second from B:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

More generally, $A_1 \times ... \times A_n$ consists of all ordered n-tuples $(x_1, ..., x_n)$ with $x_i \in A_i$ for each i.

12. Set operations and identities

October 5

Intersection The set of elements belonging to both A and B:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Thus $x \in A \cap B \leftrightarrow (x \in A \wedge x \in B)$.

Union The set of elements belonging to at least one of A and B:

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

An element belonging to both sets appears only once in their union.

Set difference The set of elements in A that are not in B, written $A \setminus B$ or $A - B$:

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}.$$

In general, $A \setminus B \neq B \setminus A$.

Universal set (universe) The set U containing all elements under consideration. Sets being discussed are subsets of U; complements are taken relative to this specified universe.

Complement The complement of A relative to U, written $\overline{A}$, consists of the elements of U that are not in A:

$$\overline{A} = U \setminus A = \{x \in U \mid x \notin A\}.$$

Changing U can change the complement, even when A stays the same.

Set identity An equality between set expressions that holds for all sets under consideration. It can be proved by showing that an arbitrary element belongs to one side if and only if it belongs to the other. Equivalently, prove that each side is a subset of the other.

Set membership table A table using 1 for membership and 0 for nonmembership. It lists all combinations of membership in the sets involved, like a truth table. Identical columns for two set expressions verify a set identity.

Venn diagram A diagram representing sets as regions, usually inside a rectangle representing U. Overlapping regions show shared elements; shading can indicate a set operation or a complement.

12.1. Named set laws

All complements below use the same universal set U.

Commutative laws for sets The order of the sets can be exchanged:

$$A \cap B = B \cap A; \quad A \cup B = B \cup A.$$

Distributive laws for sets Each of intersection and union distributes over the other:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

De Morgan's laws for sets Taking a complement exchanges intersection with union and complements each set:

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

These follow from the corresponding logical laws applied to membership.

Cardinality of a union (two-set inclusion–exclusion) For finite sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Subtract the intersection once because its elements were counted twice. Equivalently, the union splits into three nonoverlapping parts:

$$|A \cup B| = |A \cap B| + |A \setminus B| + |B \setminus A|.$$

13. Functions

October 5 and 7

Function A rule $f : A \to B$ assigning exactly one element of B to each element of A. Every input in A must have an output, and no input can have two different outputs. Different inputs may have the same output.

Domain of a function In $f : A \to B$, the set A of all permitted inputs. It is part of the specification of the function.

Codomain In $f : A \to B$, the specified set B in which the outputs must lie. Some elements of B may never occur as outputs.

Function value (image of an element) If f assigns y to the input x, write $f(x) = y$. The element y is the image of x under f.

Pre-image of an element An input x satisfying $f(x) = y$ is a pre-image of y. An element of the codomain may have no pre-image, one pre-image, or several pre-images.

Image of a set For $S \subseteq A$, the set of outputs obtained from inputs in S:

$$f(S) = \{f(a) \mid a \in S\} = \{y \in B \mid \exists a \in S, f(a) = y\}.$$

Range The set of actual outputs of f, namely $f(A)$, where A is its domain. Always $f(A) \subseteq B$; the range need not equal the codomain.

13.1. Types of functions named in the notes

The October 5 notes introduce these names; October 7 gives their defining properties and negations. Let $f : A \to B$ throughout.

Onto function (surjection) A function whose range equals its codomain. Every $y \in B$ has at least one pre-image in A:

$$\forall y \in B, \exists x \in A, f(x) = y.$$

One-to-one function (injection) A function that sends different inputs to different outputs. Equivalently,

$$\forall x_1, x_2 \in A, (f(x_1) = f(x_2) \to x_1 = x_2).$$

Equivalently, $x_1 \neq x_2 \to f(x_1) \neq f(x_2)$ for all $x_1, x_2 \in A$. Each element of the codomain has at most one pre-image.

One-to-one correspondence (bijection) A function that is both one-to-one and onto. Each element of the codomain has exactly one pre-image in the domain.

$$\forall y \in B, \exists! x \in A, f(x) = y.$$

Not injective Two distinct inputs have the same output:

$$\exists x_1, x_2 \in A, (x_1 \neq x_2 \wedge f(x_1) = f(x_2)).$$

One such pair disproves injectivity.

Not surjective Some element of the codomain has no pre-image:

$$\exists y \in B, \forall x \in A, f(x) \neq y.$$

One such element disproves surjectivity.

Not bijective The function is not injective or is not surjective (possibly both). Either of the preceding counterexamples suffices.

13.2. Combining functions

October 7

Sum of functions For functions $f, g : A \to B$ with addition defined in B and all resulting sums in B, their sum $f + g : A \to B$ is defined by

$$(f + g)(x) = f(x) + g(x).$$

Product of functions For functions $f, g : A \to B$ with multiplication defined in B and all resulting products in B, their product $fg : A \to B$ is defined by

$$(fg)(x) = f(x)g(x).$$

This multiplies the outputs at the same input.

Composition If $f : A \to B$ and $g : B \to C$, the composite function $g \circ f : A \to C$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

Apply f first, then g. More generally, composition is defined whenever the range of the inner function lies in the domain of the outer function. Composition need not commute: $g \circ f$ and $f \circ g$ can differ, and one may be defined when the other is not.

Inverse function If $f : A \to B$ is a bijection, its inverse $f^{-1} : B \to A$ sends each $y \in B$ to its unique preimage:

$$f^{-1}(y) = x \leftrightarrow f(x) = y.$$

An inverse on the entire codomain exists if and only if f is bijective. The inverse reverses the input–output assignment; f^{-1} does not mean the reciprocal $\frac{1}{f}$.

Inverse identities For a bijection $f : A \to B$,

$$\forall x \in A, f^{-1}(f(x)) = x,$$

$$\forall y \in B, f(f^{-1}(y)) = y.$$

For example, $f : \mathbb{R}^+ \to \mathbb{R}^+$ with $f(x) = x^2$ has inverse $f^{-1}(y) = \sqrt{y}$. The domain and codomain matter: the rule x^2 on $\mathbb{R} \to \mathbb{R}$ is neither injective nor surjective.

14. Sequences and recursive definitions

October 7 and 9

Sequence A function whose domain is a subset of $\mathbb{Z}$, usually $\mathbb{N}$ or $\mathbb{Z}^+$. A sequence is often written $\{a_n\}$, with its index set specified when needed. Its values can be listed in index order and need not be numbers.

Index and term The index n is an input to a sequence; the term a_n is its output at that index. Subscript notation a_n plays the role of function notation $a(n)$. Check whether indexing begins at 0 or 1.

Arithmetic sequence A numerical sequence with constant difference d between consecutive terms. With initial term $a_0 = a$,

$$a_n = a + dn.$$

Here a is the initial value and d is the common difference. With indexing starting at 1 instead, $a_n = a_1 + (n-1)d$.

Geometric sequence A numerical sequence in which each term is r times the preceding term. With initial term $b_0 = a$,

$$b_n = ar^n.$$

Here r is the common ratio. With indexing starting at 1 instead, $b_n = b_1 r^{n-1}$. The initial term is specified separately when $r = 0$, avoiding any need to evaluate 0^0.

Recursive definition A definition giving initial values and a rule for obtaining later terms from earlier terms. For example,

$$a_0 = 1; \quad a_{n+1} = \sum_{i=0}^{n} a_i \quad (n \geq 0)$$

gives the sequence $1, 1, 2, 4, 8, \ldots$.

Recurrence relation An equation relating terms of a sequence to earlier terms. Initial conditions specify the starting values; together with the recurrence, they determine the sequence under discussion.

Fibonacci sequence The sequence defined recursively by

$$f_0 = 0; \quad f_1 = 1; \quad f_{n+2} = f_n + f_{n+1} \quad (n \geq 0).$$

Its first terms are $0, 1, 1, 2, 3, 5, 8, 13, \ldots$.

Explicit formula (closed form) A formula giving a term directly from its index without computing earlier terms. The October 9 notes give the following formula for the Fibonacci sequence, to be studied later:

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right).$$

Equivalently, for $n \geq 0$,

$$f_n = \left\lfloor \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{1}{2} \right\rfloor.$$

Here $\lfloor t \rfloor$ is the greatest integer less than or equal to t.

15. Summations

October 9

Summation notation For integers $m \leq n$,

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + \ldots + a_n.$$

The lower limit m is the starting index, the upper limit n is the ending index, and both endpoints are included. The expression a_i being added is the summand.

Dummy variable (index of summation) The index bound by a summation symbol, such as i in $\sum_{i=m}^{n} a_i$. Renaming it consistently does not change the sum: $\sum_{i=m}^{n} a_i = \sum_{j=m}^{n} a_j$.

Empty sum A sum with no terms, assigned the value 0. In particular, $\sum_{i=m}^{n} a_i = 0$ when $m > n$ under the convention in the notes.

15.1. Summation rules and formulas

The following formulas use nonnegative integer n.

Linearity of summation Constants can be factored out and sums can be split term by term:

$$\sum_{k=m}^{n} (ca_k + db_k) = c \sum_{k=m}^{n} a_k + d \sum_{k=m}^{n} b_k.$$

Changing the lower limit by subtraction For $1 \le m \le n$,

$$\sum_{k=m}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{m-1} a_k.$$

This removes the unwanted initial terms from a known sum.

Sum of consecutive integers

$$\sum_{k=0}^{n} k = \sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

Finite geometric sum For $r \neq 1$,

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}.$$

If $r = 1$, the sum is $n + 1$. With initial term a, multiply the right-hand side by a to sum $a + ar + ... + ar^n$.

Sum of squares

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Sum of cubes

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

16. Cardinalities of finite and infinite sets

October 9

Same cardinality (equinumerous sets) Sets S and T have the same cardinality, written $|S| = |T|$, if there exists a bijection $f : S \to T$. This extends the finite notion of having the same number of elements to infinite sets.

Cardinality comparison $|S| \le |T|$ means that there exists an injection $f : S \to T$. Thus distinct elements of S can be assigned distinct elements of T.

Transitivity of cardinality equality If $|S| = |T|$ and $|T| = |R|$, then $|S| = |R|$. Composing the two bijections gives a bijection from S to R.

Transitivity of cardinality comparison If $|S| \le |T|$ and $|T| \le |R|$, then $|S| \le |R|$. Composing the two injections gives an injection from S to R.

Schröder–Bernstein theorem If $|S| \le |T|$ and $|T| \le |S|$, then $|S| = |T|$. Injections in both directions guarantee that a bijection exists. The theorem is stated without proof in the October 9 notes.

The infinite-set equalities stated in that lecture are

$$|\mathbb{N}| = |\mathbb{Z}^+| = |\mathbb{Z}|; \quad |\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|.$$

An infinite set can have the same cardinality as a proper subset of itself; for example, $n \mapsto n + 1$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}^+$.

17. Corrections applied in this glossary

- September 16: the inverse of $p \to q$ is $\neg p \to \neg q$. An implication excludes p true with q false.
- September 21: negating an existential quantifier gives $\forall x(\neg P(x))$. Existential quantification over a finite domain expands into a disjunction, rather than an arithmetic sum.
- September 23: $\neg(p \to q) \equiv p \wedge \neg q$. Negating a strict bound $|f(x) - f(x_0)| < \varepsilon$ gives $|f(x) - f(x_0)| \geq \varepsilon$.
- September 25: the usual "affirming the conclusion" fallacy includes both premises $p \to q$ and q.
- October 5: membership in an intersection is $x \in A \cap B \leftrightarrow (x \in A \wedge x \in B)$. Set expressions use $\cup$ for union, so the distributive example begins with $(A \cap B) \cup C$. A function assigns exactly one output to each input; an output can have several pre-images.
- October 7: the sum of functions is defined by $(f + g)(x) = f(x) + g(x)$. Composition requires compatible inputs and outputs; unequal domain and codomain sets alone do not rule it out. The second inverse identity is $f(f^{-1}(y)) = y$ for $y \in B$. For the square function on all real inputs, use $\mathbb{R}$ (or the nonnegative reals) as codomain, since $0^2 = 0$ is not a positive real number.
- October 9: the geometric-sum formula uses one consistent upper index n and exponent $n + 1$; its $r = 1$ case has $n + 1$ terms. The sum-of-cubes formula is $\left(\frac{n(n+1)}{2}\right)^2$. In the lower-limit example, the correct expansion is

$$\sum_{k=9}^{15}(3k^3 - 7k^2) = 3\sum_{k=1}^{15}k^3 - 3\sum_{k=1}^{8}k^3 - 7\sum_{k=1}^{15}k^2 + 7\sum_{k=1}^{8}k^2.$$

  For a finite set of size k, the counting bijection uses $\{1, \ldots, k\}$ (the empty set if $k = 0$), rather than always $\{1, 2, 3, 4, 5\}$. The cardinality theorem is named Schröder–Bernstein.